

Experiment 3

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.MODEL SMALL

.STACK 100H

.DATA

NUM DB 5 ; Number for factorial (change this value as needed)

FACT DW 1 ; Variable to store factorial result

MSG DB 'Factorial: \$' ; Message to display before result

.CODE

MAIN PROC

MOV AX, @DATA

MOV DS, AX

MOV AL, NUM ; Load number in AL

CBW ; Convert AL to AX (sign-extend)

CALL FACTORIAL ; Call factorial procedure

MOV DX, OFFSET MSG

MOV AH, 09H

INT 21H ; Print message

CALL PRINT_NUM ; Print the factorial result

MOV AH, 4CH

INT 21H ; Exit program

MAIN ENDP

; Factorial Procedure

FACTORIAL PROC

MOV CX, AX ; Move number to CX for loop counter

MOV AX, 1 ; Initialize AX = 1 (Factorial starts at 1)

FACTORIAL_LOOP:

MUL CX ; AX = AX * CX

LOOP FACTORIAL_LOOP

MOV FACT, AX ; Store the result in FACT

RET

FACTORIAL ENDP

; Print Number Procedure

PRINT_NUM PROC

MOV AX, FACT ; Load factorial result

MOV CX, 0 ; Clear CX (digit counter)

NEXT_DIGIT:

MOV DX, 0

MOV BX, 10

DIV BX ; AX / 10 → Quotient in AX, Remainder in DX

PUSH DX ; Push remainder (digit) onto stack

INC CX ; Increment digit counter

TEST AX, AX ; Check if AX is zero

JNZ NEXT_DIGIT ; If not, continue extracting digits

PRINT_LOOP:

POP DX ; Get digit from stack

ADD DL, '0' ; Convert to ASCII

MOV AH, 02H

INT 21H ; Print digit

LOOP PRINT_LOOP ; Repeat for remaining digits

RET

PRINT_NUM ENDP

END MAIN

OUTPUT: